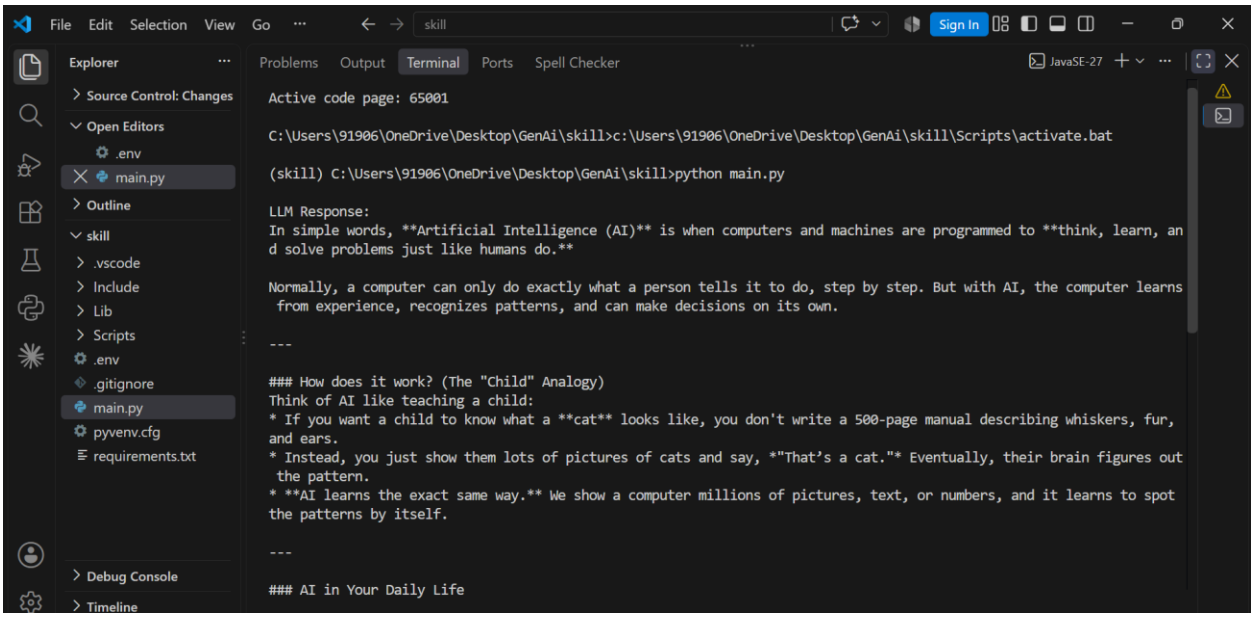
GENERATIVE AI AND AI PLATFORM ENGINEERING

Course Code: 25CS2207E

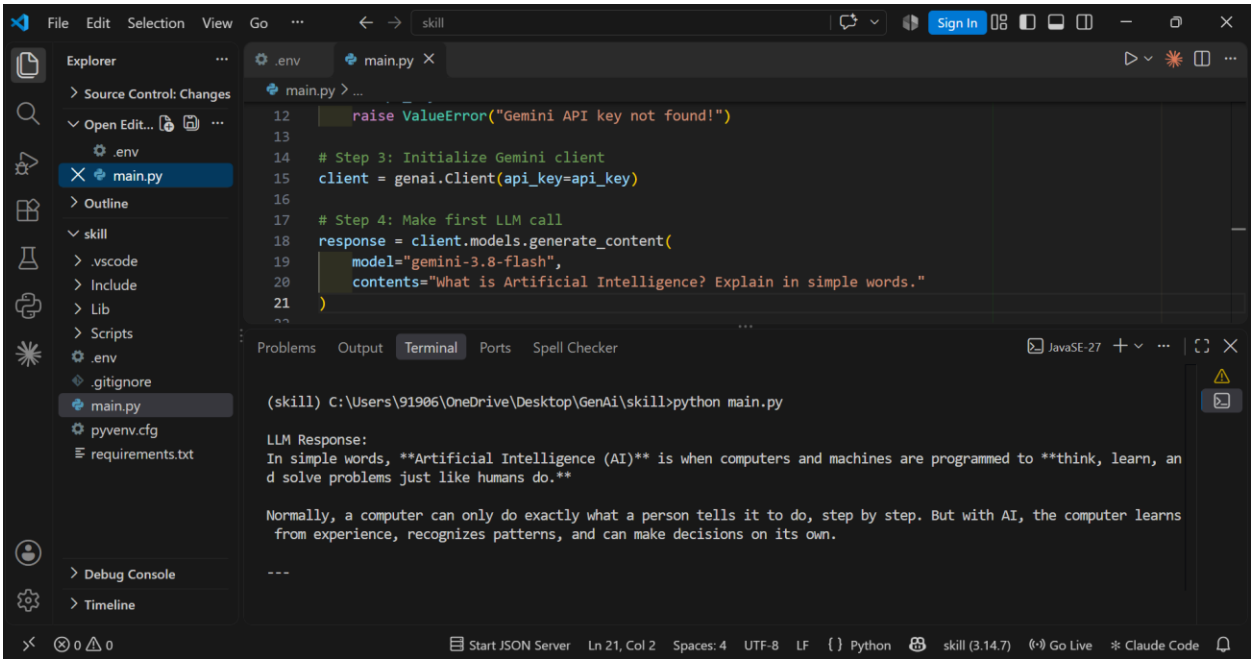
Topic - Week 01 VS Code output

Name : S. Manvi Vaishnavi
Roll Number : 2520030326



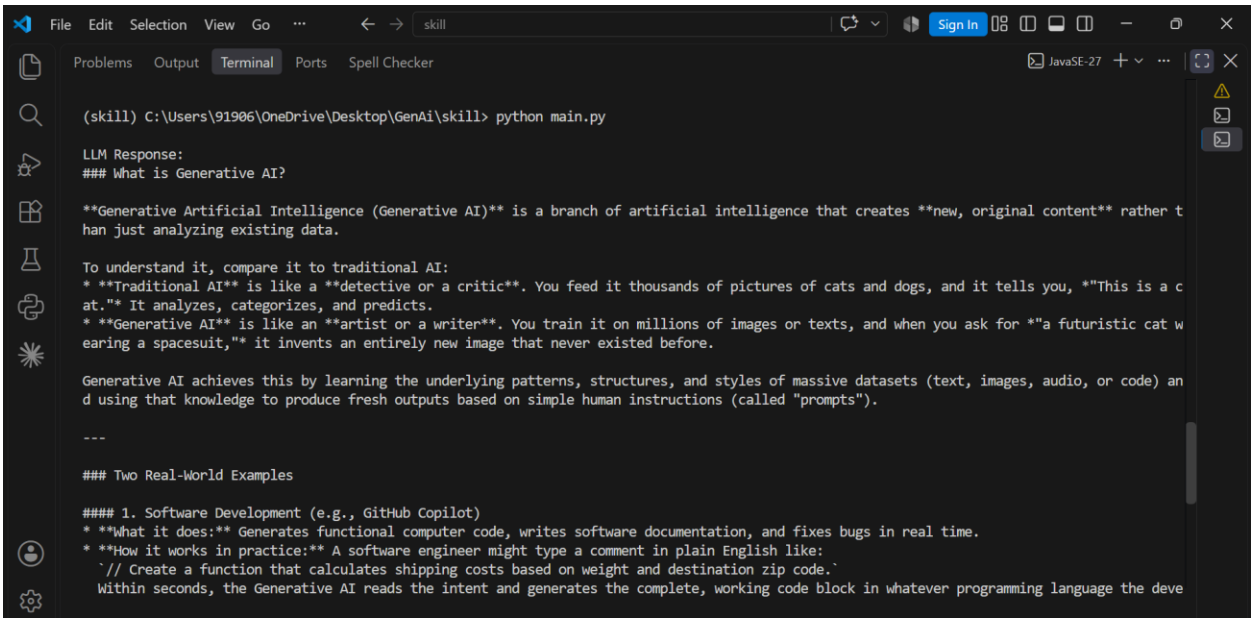
1.contents="What is Artificial Intelligence? Explain in simple words."

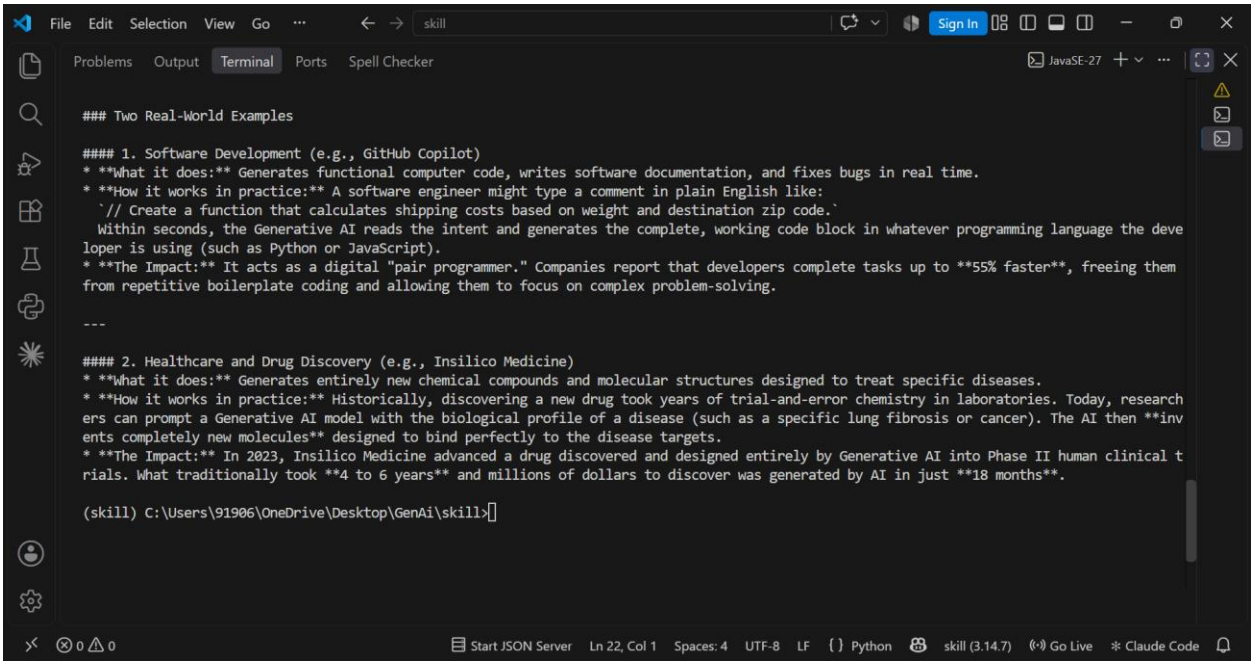
OUTPUT:



2. contents="Explain Generative AI with two real-world examples."

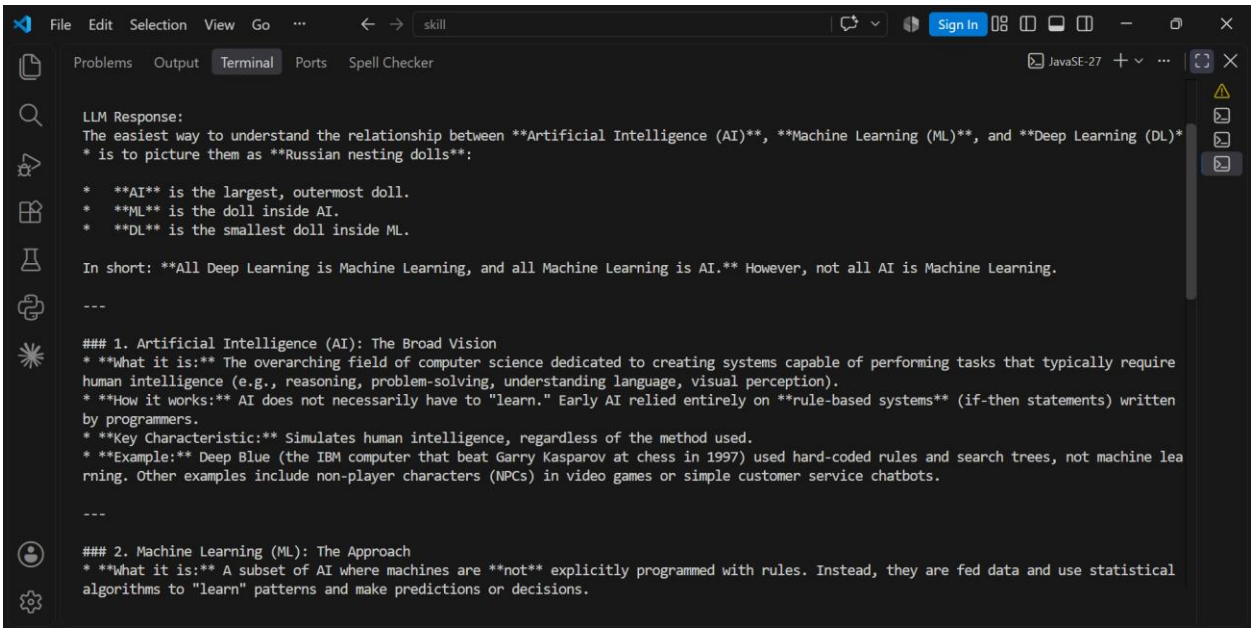
OUTPUT:

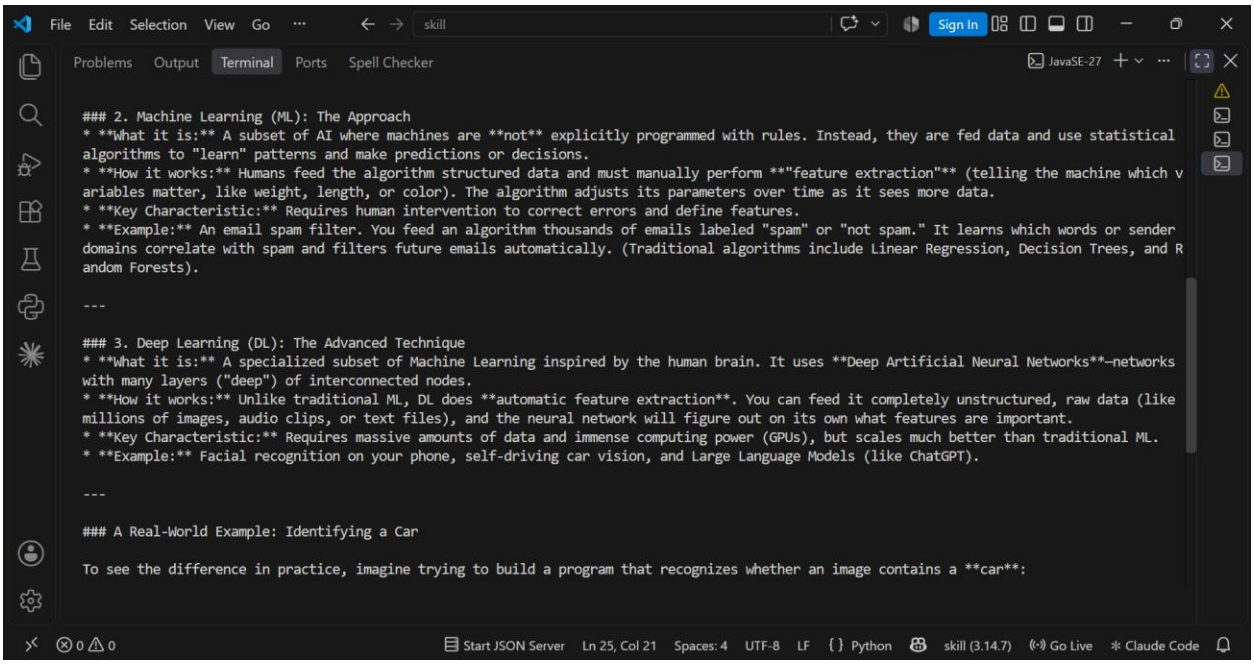


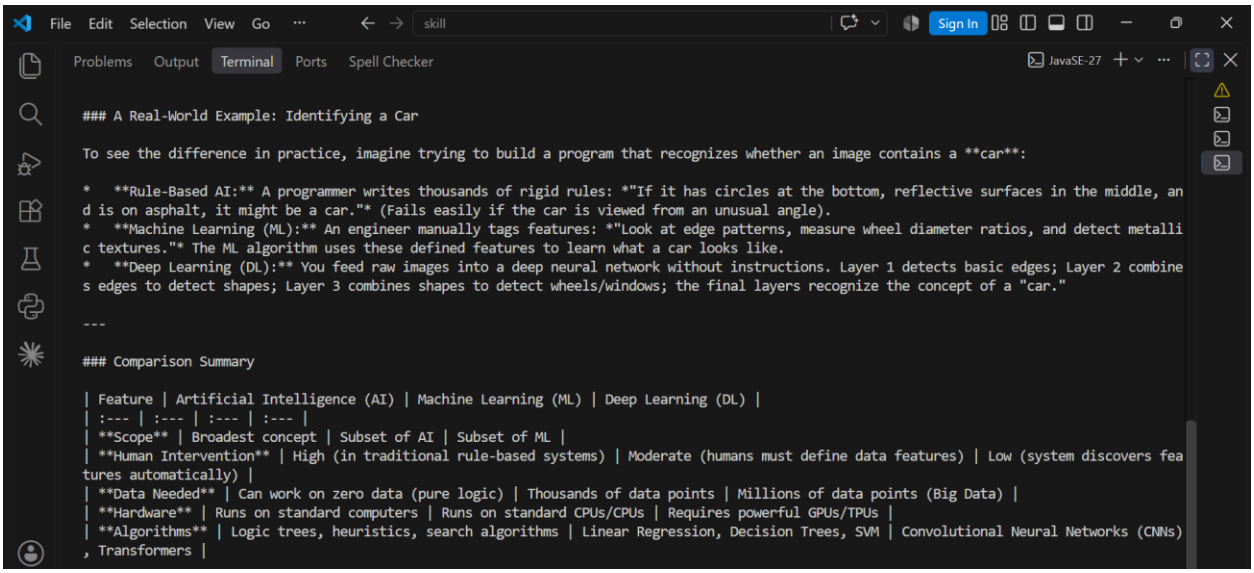


3. "Differentiate Artificial Intelligence, Machine Learning, and Deep Learning."

OUTPUT:

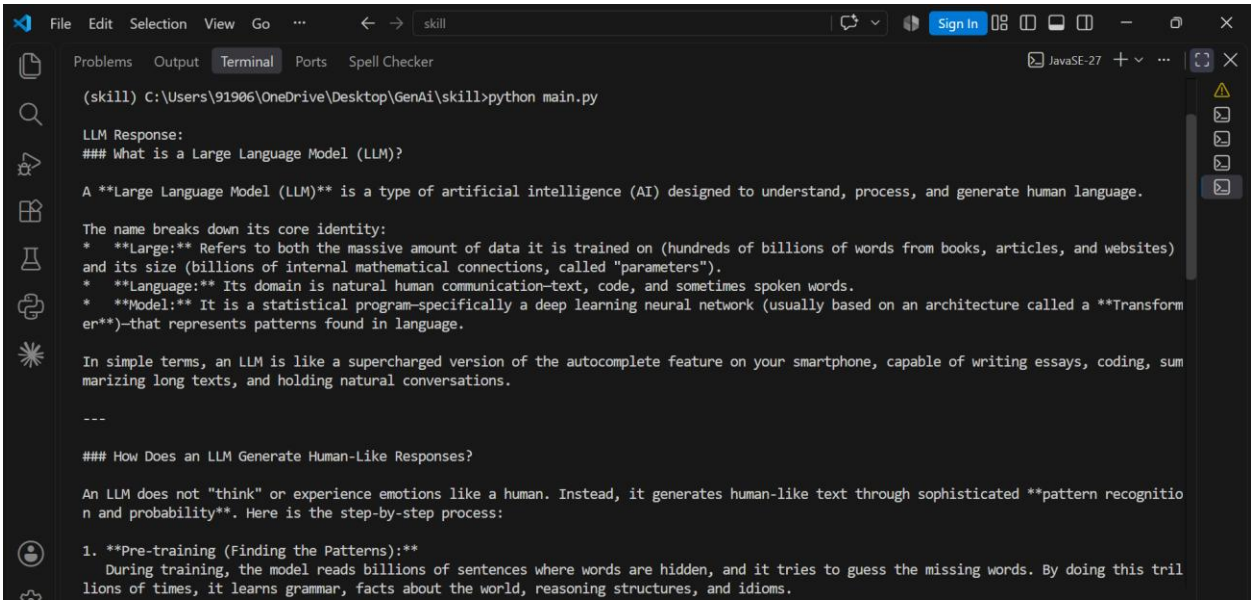






4. Explain what a Large Language Model (LLM) is and how it generates human-like responses. Give two examples.

OUTPUT:




```
(skill) C:\Users\91906\OneDrive\Desktop\GenAi\skill>python main.py

LLM Response:
### What is a Large Language Model (LLM)?

A **Large Language Model (LLM)** is a type of artificial intelligence (AI) designed to understand, process, and generate human language.

The name breaks down its core identity:
*   **Large:** Refers to both the massive amount of data it is trained on (hundreds of billions of words from books, articles, and websites) and its size (billions of internal mathematical connections, called "parameters").
*   **Language:** Its domain is natural human communication—text, code, and sometimes spoken words.
*   **Model:** It is a statistical program—specifically a deep learning neural network (usually based on an architecture called a **Transformer**)—that represents patterns found in language.

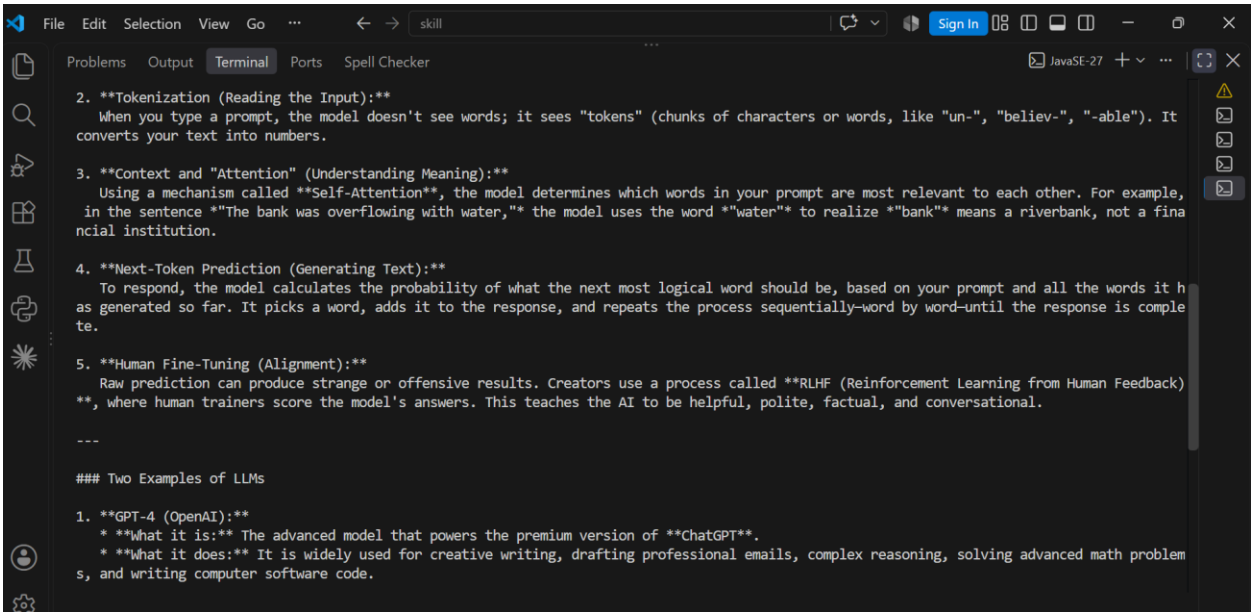
In simple terms, an LLM is like a supercharged version of the autocomplete feature on your smartphone, capable of writing essays, coding, summarizing long texts, and holding natural conversations.

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### How Does an LLM Generate Human-Like Responses?

An LLM does not "think" or experience emotions like a human. Instead, it generates human-like text through sophisticated **pattern recognition and probability**. Here is the step-by-step process:

1. **Pre-training (Finding the Patterns):**
   During training, the model reads billions of sentences where words are hidden, and it tries to guess the missing words. By doing this trillions of times, it learns grammar, facts about the world, reasoning structures, and idioms.
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2. **Tokenization (Reading the Input):**
   When you type a prompt, the model doesn't see words; it sees "tokens" (chunks of characters or words, like "un-", "believ-", "-able"). It converts your text into numbers.

3. **Context and "Attention" (Understanding Meaning):**
   Using a mechanism called **Self-Attention**, the model determines which words in your prompt are most relevant to each other. For example, in the sentence *"The bank was overflowing with water,"* the model uses the word *"water"* to realize *"bank"* means a riverbank, not a financial institution.

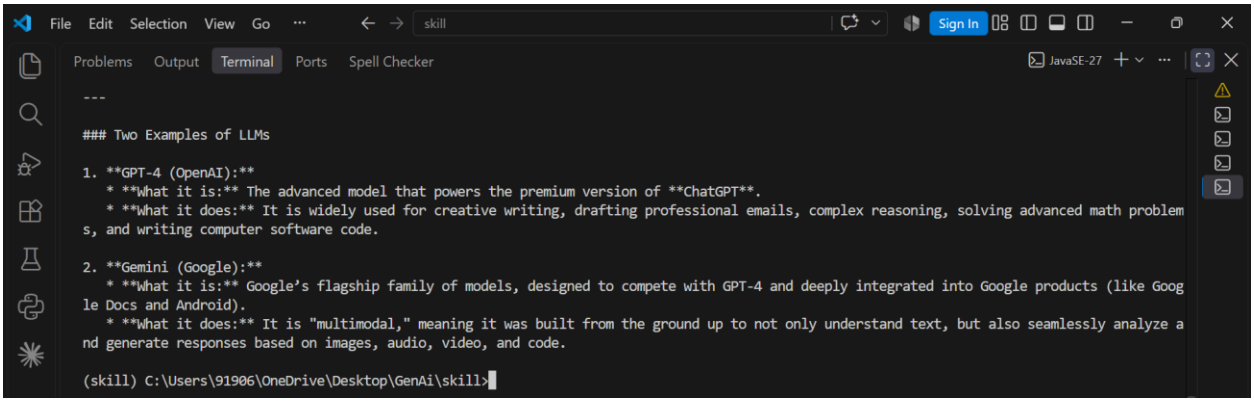
4. **Next-Token Prediction (Generating Text):**
   To respond, the model calculates the probability of what the next most logical word should be, based on your prompt and all the words it has generated so far. It picks a word, adds it to the response, and repeats the process sequentially—word by word—until the response is complete.

5. **Human Fine-Tuning (Alignment):**
   Raw prediction can produce strange or offensive results. Creators use a process called **RLHF (Reinforcement Learning from Human Feedback)**, where human trainers score the model's answers. This teaches the AI to be helpful, polite, factual, and conversational.

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### Two Examples of LLMs

1. **GPT-4 (OpenAI):**
   * **What it is:** The advanced model that powers the premium version of **ChatGPT**.
   * **What it does:** It is widely used for creative writing, drafting professional emails, complex reasoning, solving advanced math problems, and writing computer software code.
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File Edit Selection View Go
skill
Sign In
Problems Output Terminal Ports Spell Checker
JavaSE-27

Two Examples of LLMs
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* **What it does:** It is widely used for creative writing, drafting professional emails, complex reasoning, solving advanced math problems, and writing computer software code.
2. **Gemini (Google):**
* **What it is:** Google's flagship family of models, designed to compete with GPT-4 and deeply integrated into Google products (like Google Docs and Android).
* **What it does:** It is "multimodal," meaning it was built from the ground up to not only understand text, but also seamlessly analyze and generate responses based on images, audio, video, and code.
(skill) C:\Users\91906\OneDrive\Desktop\GenAi\skill>